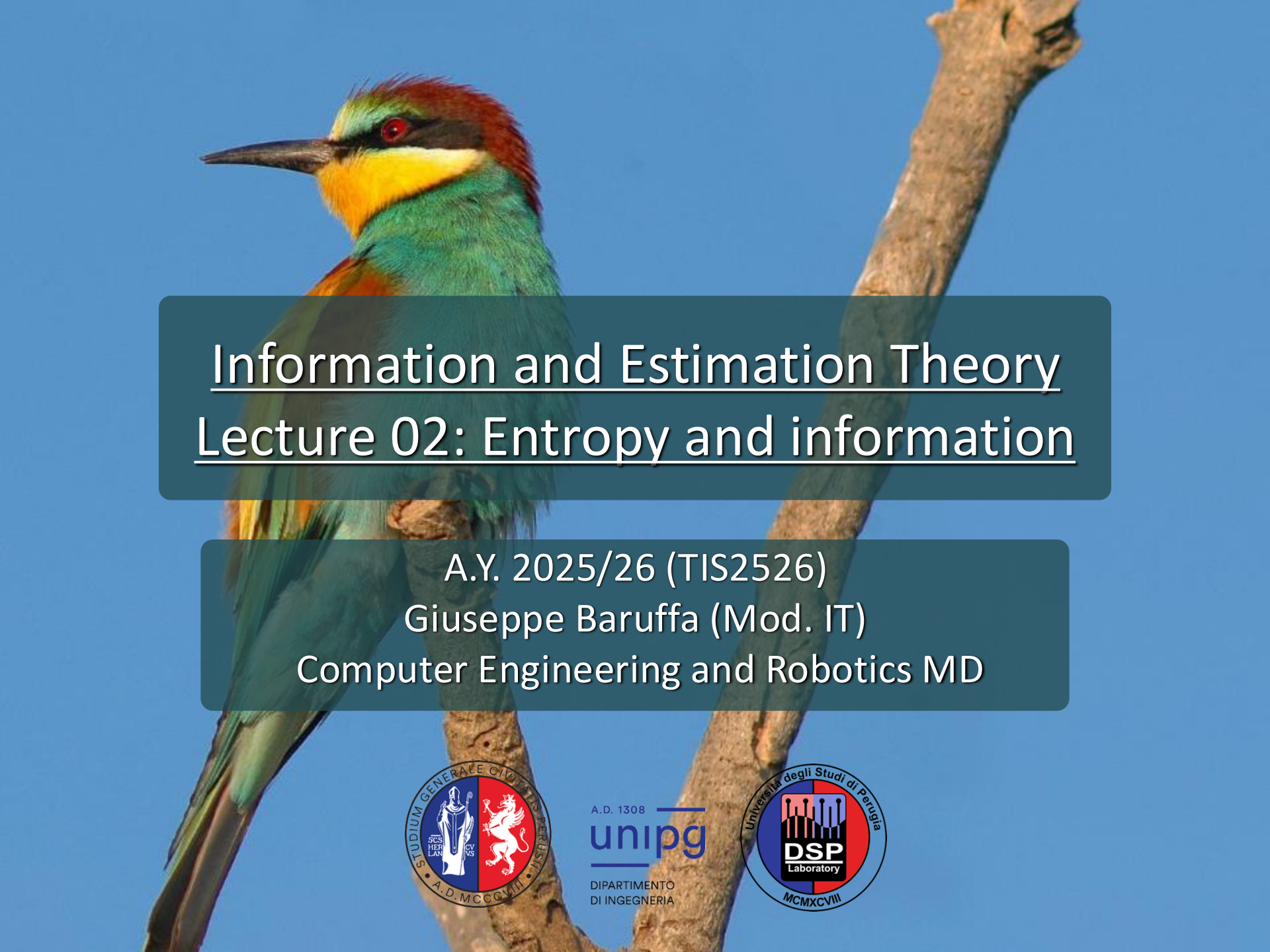


Information and Estimation Theory
Lecture 02: Entropy and information

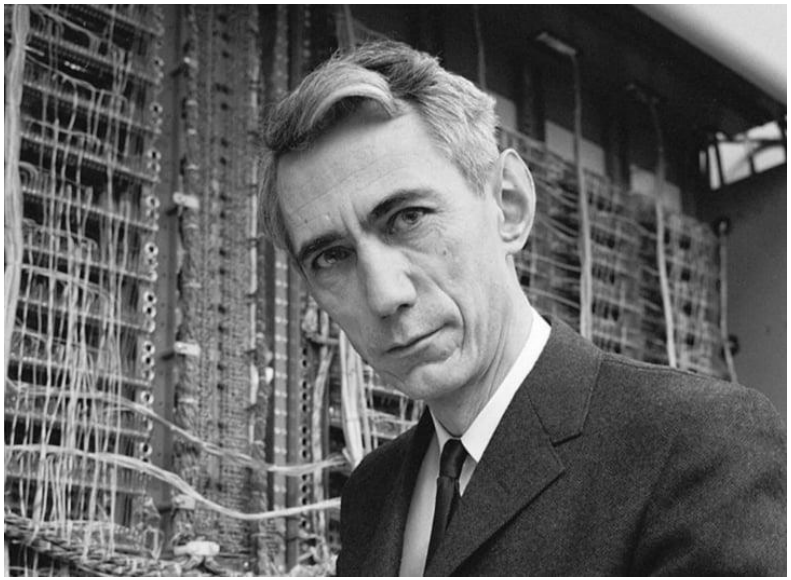
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Measuring Information

- Need for a «meter» to weigh the amount of information contained in an event, a message, etc.
- **Claude Shannon**, in 1948, introduced a method based on the probability of occurrence of events
- This probability measures the information carried by a specific message or event
  - **High probability** → <u>low information</u>
    - "Tomorrow the sun will rise"
      - ≈0.9999999999994524
  - **Low probability** → <u>high information</u>
    - "Tomorrow gas stations are closed"
      - ≈0.0010267
- The calculated measure of uncertainty allows us to make careful decisions



Calculating the Information

- Consider a discrete source that emits random letters (chosen from a finite alphabet) with a probability *p*

$$\begin{array}{lll} A & \rightarrow & p_A \\ B & \rightarrow & p_B \\ \mathrm{M} & \mathrm{M} & \mathrm{M} \\ Z & \rightarrow & p_Z \end{array}$$

- In a message of *N* letters

$$\begin{array}{llll} A & \text{for} & \approx Np_A & \text{times} \\ B & \text{for} & \approx Np_B & \text{times} \\ \vdots & \vdots & \vdots & \vdots \\ Z & \text{for} & \approx Np_Z & \text{times} \end{array}$$

- For Shannon, the information given by each letter of the alphabet is

$$I(A) = \log_2 \frac{1}{p_A}, \qquad I(B) = \log_2 \frac{1}{p_B}, \qquad \ldots [\text{bit}]$$

- And the average total information is

$$I_{\mathrm{TOT}} = p_A N \log_2 \frac{1}{p_A} + p_B N \log_2 \frac{1}{p_B} + \cdots + p_Z N \log_2 \frac{1}{p_Z}$$

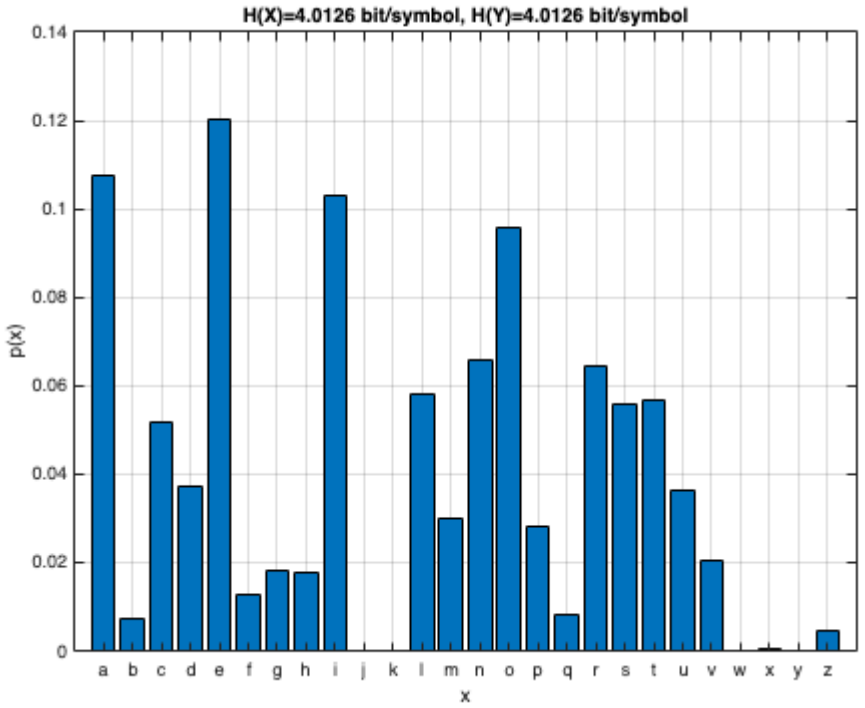
Bits and Entropy

- The average information, for each letter, is thus

$$H = \frac{I_{\mathrm{TOT}}}{N} = p_A \log_2 \frac{1}{p_A} + p_B \log_2 \frac{1}{p_B} + \ldots + p_Z \log_2 \frac{1}{p_Z} \qquad \left[\frac{\mathrm{bit}}{\mathrm{letter}}\right]$$

- ***H*** is called **entropy** (term borrowed from chemistry and physics), it measures the average capability of a source to give information at each emitted letter
- For instance, by applying this formula to **Divina Commedia** (400k letters with the estimated probabilities here) we obtain

$$H = 4.0126 \ \ \mathrm{bit/letter}$$



Formal Definition of Entropy

- Entropy measures the **uncertainty** of a random variable (RV)
- Let X be a discrete RV with alphabet $\mathcal{X}$ and probability mass function (PMF) $p(x) = \Pr\{X = x\}$, $x \in \mathcal{X}$
- The PMF $p(x)$ denotes $f_X(x)$: thus, $p(x)$ and $p(y)$ refer to two different RVs and PMFs
- The **entropy $H(X)$** or **$H(p)$** of a discrete RV X is

$$H(X) = -\sum_{x \in \mathcal{X}} p(x) \log p(x)$$

0 log(0) = 0 for continuity

$$\lim_{p \to 0^+} p \log p = \lim_{p \to 0^+} \frac{\log p}{1/p} = \lim_{p \to 0^+} \frac{1/p}{-1/p^2} = \lim_{p \to 0^+} (-p) = 0$$

or choose $\mathcal{X} = \{x : p(x) > 0\}$

- If the logarithm is base b, we denote the entropy as $H_b(X)$
  - Unless otherwise specified, b=2, and entropy is measured in *bits*
  - If b=e, it is measured in *nats*; when b=10, we have *dits* (or *Hartleys*)

$\ln(\cdot) = \log_e(\cdot)$
$\log(\cdot) = \log_2(\cdot)$

- Note that entropy is a <u>functional</u> of the distribution of X
  - It does not depend on the values taken by X, but only on the probabilities

Entropy as Expected Value

- If $X \sim p(x)$, the **expected value $\mathrm{E}_p\{g(X)\}$** of the RV $g(X)$ is

$$\mathrm{E}_p\{g(X)\} = \sum_{x \in \mathcal{X}} g(x)p(x)$$

- The entropy of X can be interpreted as the expected value of the RV $g(X) = \log[1/p(X)]$, where X is drawn according to PMF $p(x)$

$$H(X) = \mathrm{E}_p\left\{\log \frac{1}{p(X)}\right\}$$

- The definition of entropy answers to natural questions such as:
  - "What is the average length of the shortest description of the RV X?"

- However, it can be derived *axiomatically* by defining certain properties that it must satisfy

Axiomatic Definitions of Entropy

- Define some axioms (due to Shannon and Kinchin) for the measure of information
- Consider a sequence of symmetric functions $H_m(p_1, \dots, p_m)$ with properties

1. Normalization → $H_2\left(\frac{1}{2}, \frac{1}{2}\right) = 1$ (or maximization)
2. Continuity → $H_2(p, 1-p)$ is continuous in p
3. Grouping → $$H_m(p_1, \dots, p_m) = H_{m-1}(p_1 + p_2, \dots, p_m) + (p_1 + p_2) H_2\left(\frac{p_1}{p_1 + p_2}, \frac{p_2}{p_1 + p_2}\right)$$

- It can be proven (difficult) that H_m **must** have the form

$$H_m(p_1, \dots, p_m) = -\sum_{i=1}^{m} p_i \log_b p_i$$

for any $m \geq 2$ and $b > 1$

Entropy Positivity and Base

- **Lemma:** Entropy is always nonnegative, $H(X) \geq 0$
  - *Proof:* $0 \leq p(x) \leq 1$ implies that $\log[1/p(x)] \geq 0$

- **Lemma:** Base change, $H_b(X) = (\log_b a)\, H_a(X)$
  - *Proof:* $log_b(\cdot) = \log_b(a)\, \log_a(\cdot)$

$$H_b(X) = \mathrm{E}_p\left\{\log_b \frac{1}{p(X)}\right\} = \mathrm{E}_p\left\{\log_b a \log_a \frac{1}{p(X)}\right\} = \log_b a\, \mathrm{E}_p\left\{\log_a \frac{1}{p(X)}\right\}$$
$$= \log_b a\, H_a(X)$$

- The second lemma enables us to change the base of the logarithm in the definition
  - Entropy can be changed from one base to another by multiplying by the appropriate factor
  - Switching between base *e* and base 2 allows simplifying some mathematical derivations

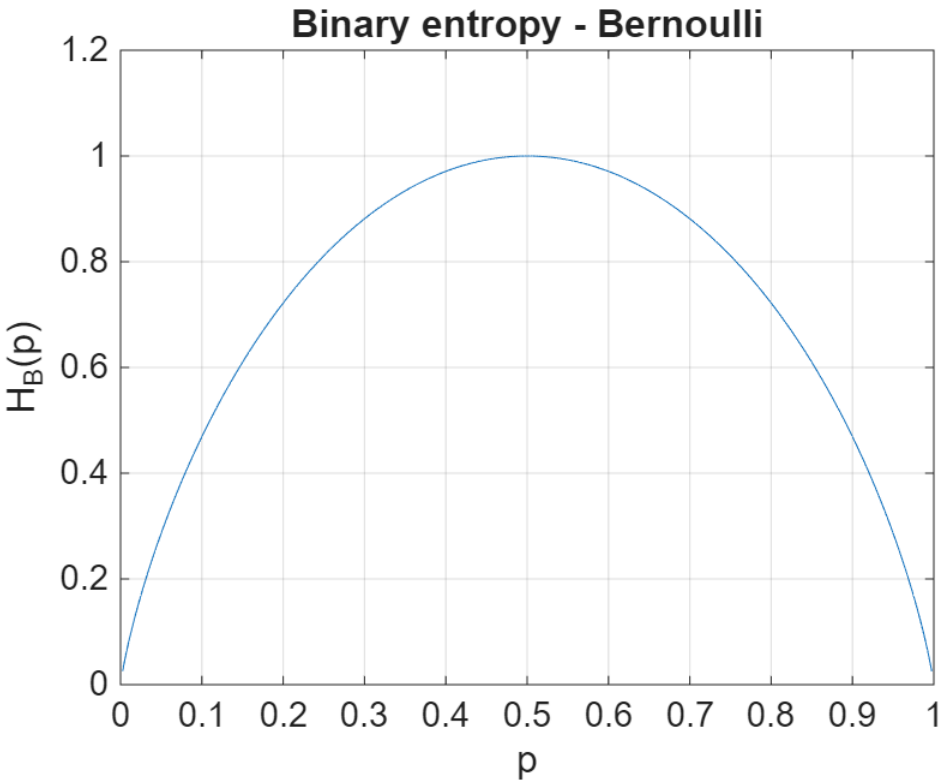
Example: Binary Entropy (Bernoulli PMF)

- Let $X \sim \text{Bern}(p), \qquad X = \begin{cases} 1, & \text{with probability } p \\ 0, & \text{with probability } 1-p \end{cases}$

Subscript B is not in italic, then it is not a base

- Then $H(X) = -p \log p - (1-p)\log(1-p) \stackrel{\text{def}}{=} H_B(p)$

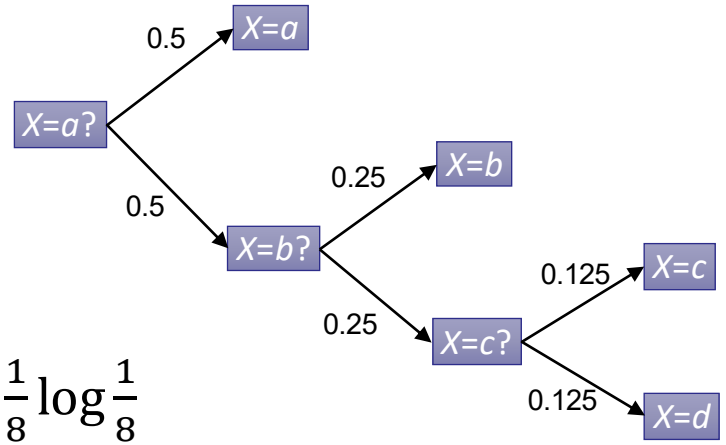
Remember $\lim_{p \to 0^+} p \log p = 0$



- $H_B(p) = 1$ bit when $p = 1/2$
- It is a **concave** function of the PMF and equals 0 when $p = 0$ or 1
  - This makes sense, because when $p = 0$ or 1, the variable is not random (no uncertainty)
- The uncertainty is maximum for $p = 1/2$, which is the maximum value of the binary entropy

Example: Entropy and Binary Questions

- Let $X = \begin{cases} a, & \text{with prob. } 1/2 \\ b, & \text{with prob. } 1/4 \\ c, & \text{with prob. } 1/8 \\ d, & \text{with prob. } 1/8 \end{cases}$



Then, $H(X) = -\frac{1}{2}\log\frac{1}{2} - \frac{1}{4}\log\frac{1}{4} - \frac{1}{8}\log\frac{1}{8} - \frac{1}{8}\log\frac{1}{8}$

$$= -\frac{1}{2}(-1) - \frac{1}{4}(-2) - 2\frac{1}{8}(-3) = \frac{7}{4} = 1.75\ bit$$

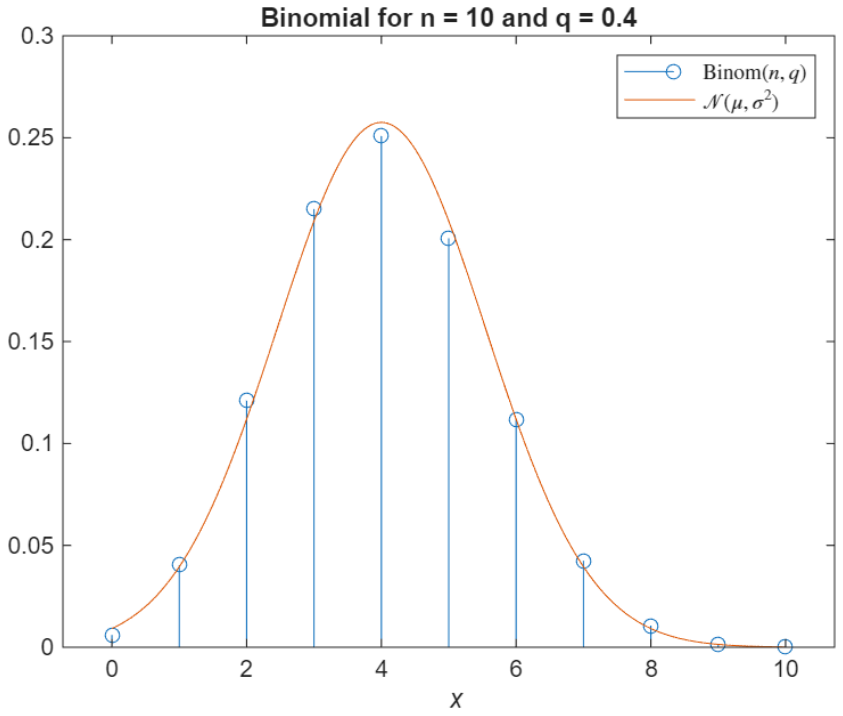
- Guess *X* with the min. number *Q* of binary "*yes/no*" questions
  - 1st question: "Is *X* = *a*?" → splits the probability in half
  - If the answer is no, the 2nd question can be "Is *X* = *b*?"
  - The 3rd question can be "Is *X* = *c*?"
- Here, the expected number of binary questions is
  - $\mathrm{E}\{Q\} = \sum_q q \Pr\{Q = q\} = 0.5 \cdot 1 + 0.25 \cdot 2 + 0.125 \cdot 3 = 1.75$
  - In general, the minimum number of binary questions approaches *H*(*X*) from above when *p*(*x*) is not an integer power of 2

Example: Entropy of Binomial Distribution

- The binomial PMF $p(x)$ of a RV $X \sim \text{Binom}(n, q)$ is the probability that x of n i.i.d. variates, distributed as $B_i \sim \text{Bern}(q)$, are 1's

$$p(x) = \Pr\{\textstyle\sum_i B_i = x\} = \binom{n}{x} q^x (1-q)^{n-x}, \quad x = 0,1,\dots,n$$

  - For large *n*, the PMF resembles that of a Gaussian, $\phi(x)$, with mean $\mu = nq$ and variance $\sigma^2 = nq(1-q)$



- Its entropy is complex to represent in closed form, but with the Gaussian approximation we can find (for *n* not too small)

$$H(X) \approx \max\left\{0, \frac{1}{2}\log 2\pi e n q(1-q)\right\}$$

Example: Entropy of Binomial Distribution

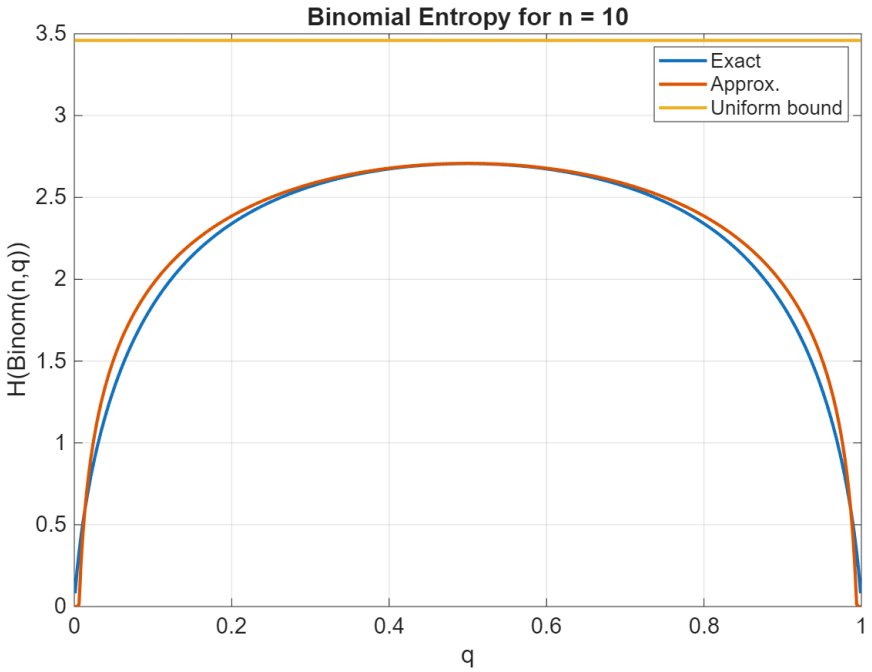
- *Proof*: use a first Gaussian approximation and calculate *nats*

$$H_e(X) = -\sum_{x=0}^{n} p(x) \ln\left(\binom{n}{x} q^x (1-q)^{n-x}\right) \approx -\sum_{x=0}^{n} \phi(x) \ln\left(\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}\right)$$

$$= -\ln\frac{1}{\sqrt{2\pi\sigma^2}} \sum_{x=0}^{n} \phi(x) + \frac{1}{2\sigma^2} \sum_{x=0}^{n} \phi(x)(x-\mu)^2 \approx \frac{1}{2}\ln 2\pi\sigma^2 + \frac{1}{2\sigma^2}\sigma^2$$

$$= \frac{1}{2}\ln 2\pi\sigma^2 + \frac{1}{2}\ln e = \frac{1}{2}\ln 2\pi enq(1-q)$$

- Second approximation for Gaussian
  - Area and 2nd moment with sums
  - Tails missing
- Apply base change to get it in *bits*
- For low or high values of q the approximation could become negative
- Also shown, the upper bound of discrete entropies



Example: Entropy of Poisson distribution

- The Poisson PMF $p(x)$ of the RV $X \sim \text{Pois}(\lambda)$ is the probability to observe x events (over a fixed interval) when λ events occur on average

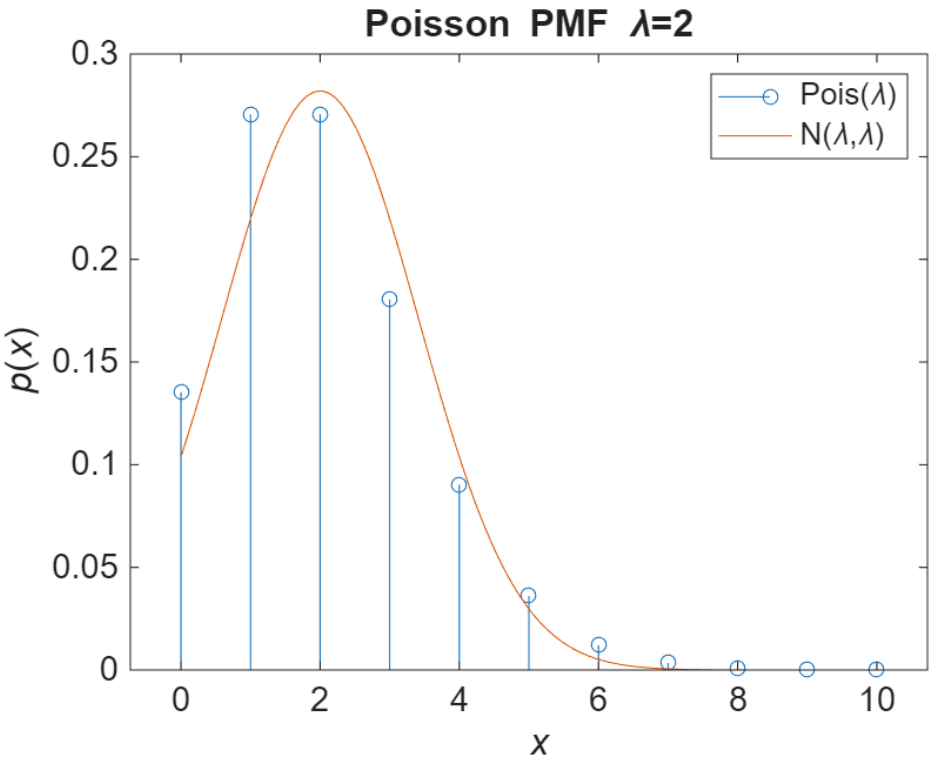
$$p(x) = \frac{\lambda^x e^{-\lambda}}{x!}, \qquad x \in \mathbb{Z}^+$$

- Its mean and variance are both λ (Gaussian-like)
  - E.g., the probability of 3 network packets in 1s, when they arrive at an average rate of 2 packets/s, is

$$p(3) = \frac{2^3 e^{-2}}{3!} = \frac{8 \cdot 0.13}{6} \approx 0.18$$

- Approximated entropy is (as for the binomial case)

$$H(X) \approx \frac{1}{2} \log 2\pi e\lambda$$



Example: Entropy of Poisson distribution

- *Proof*: coarse approximation with Stirling and upper bound

$$H_e(X) = -\sum_{x=0}^{\infty} p(x) \ln \frac{\lambda^x e^{-\lambda}}{x!} = -\sum_{x=0}^{\infty} p(x)(x \ln \lambda - \lambda - \ln x!)$$

$$= -\ln \lambda \sum_{x=0}^{\infty} p(x)x + \lambda \sum_{x=0}^{\infty} p(x) + \sum_{x=0}^{\infty} p(x) \ln x! = \lambda(1 - \ln \lambda) + \sum_{x=0}^{\infty} p(x) \ln x!$$

Stirling's approximation less O(log x)

$$\approx \lambda(1 - \ln \lambda) + \sum_{x=0}^{\infty} p(x)(x \ln x - x)$$

MacLaurin expansion of the exponential

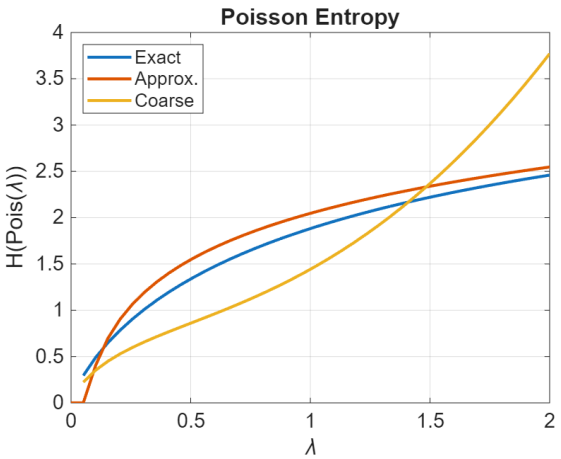
$$= \lambda(1 - \ln \lambda) + e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x \ln x}{(x-1)!} - e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x}{(x-1)!}$$

$$= \lambda(1 - \ln \lambda) + e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x \ln x}{(x-1)!} - e^{-\lambda} e^{\lambda} \lambda$$

$$\approx -\lambda \ln \lambda + e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x (x-1)}{(x-1)!}$$

$\ln(x) \le x - 1$, since in $x = 1$ the tangent of $\ln(x)$ is exactly $(x - 1)$

$$= -\lambda \ln \lambda + e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x}{(x-2)!} = -\lambda \ln \lambda + e^{-\lambda} e^{\lambda} \lambda^2 \le \lambda(\lambda - \ln \lambda)$$



Joint Entropy

- We now extend the definition to a pair of RVs *X* and *Y*
- *N*ote that (*X*,*Y*) can be considered as a single, vector-valued RV

- The **joint entropy *H*(*X*,*Y*)** of a pair of discrete RV (*X*,*Y*) with a joint PMF *p*(*x*,*y*) is

$$H(X,Y) = -\sum_{x\in\mathcal{X}}\sum_{y\in\mathcal{Y}} p(x,y)\log p(x,y)$$

written also as $H(X,Y) = -\mathrm{E}\{\log p(X,Y)\}$

Conditional Entropy

- If $(X, Y) \sim p(x, y)$, we define the **conditional entropy $\boldsymbol{H(Y|X)}$** of a RV *Y* given another *X* as the expected value of the entropies of the conditional PMFs of *Y*, averaged over the conditioning RV *X*

$$H(Y|X) = \sum_{x \in \mathcal{X}} p(x) H(Y|X = x) = -\sum_{x \in \mathcal{X}} p(x) \sum_{y \in \mathcal{Y}} p(y|x) \log p(y|x)$$

$$= -\sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x) p(y|x) \log p(y|x) = -\sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log p(y|x)$$

$$= -\mathrm{E}\{\log p(Y|X)\}$$

  - Remember that $p(x, y) = p(x) p(y|x)$

Chain Rule Theorem (and Corollary)

- **Theorem:** the joint entropy of a pair of RVs is the entropy of one plus the conditional entropy of the other

$$H(X,Y) = H(X) + H(Y|X)$$

  - *Proof:*

$$H(X,Y) = -\sum_{x\in\mathcal{X}}\sum_{y\in\mathcal{Y}} p(x,y)\log p(x,y) = -\sum_{x\in\mathcal{X}}\sum_{y\in\mathcal{Y}} p(x,y)\log p(x)p(y|x)$$

$$= -\sum_{x\in\mathcal{X}}\sum_{y\in\mathcal{Y}} p(x,y)\log p(x) - \sum_{x\in\mathcal{X}}\sum_{y\in\mathcal{Y}} p(x,y)\log p(y|x)$$

$$= -\sum_{x\in\mathcal{X}} p(x)\log p(x) - \sum_{x\in\mathcal{X}}\sum_{y\in\mathcal{Y}} p(x,y)\log p(y|x) = H(X) + H(Y|X)$$

  - Or use $\log p(X,Y) = \log p(X) + \log p(Y|X)$ and expect both sides

- **Corollary:** $H(X,Y|Z) = H(X|Z) + H(Y|X,Z)$
  - *Proof:* follows the same lines of the theorem

Example: Entropies of Joint PMF

- Let (X, Y) have the joint PMF →
- The marginal PMF of X is (1/2,1/4,1/8,1/8) and the marginal PMF of Y is (1/4,1/4,1/4,1/4)
  - $H(X)$=7/4=1.75 bits and $H(Y)$=2 bits

Y \ X	1	2	3	4
1	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$	$\frac{1}{32}$
2	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{32}$	$\frac{1}{32}$
3	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$
4	$\frac{1}{4}$	0	0	0

$p(x)$ = (1/2, 1/4, 1/8, 1/8)

(1/4, 1/4, 1/4, 1/4) = $p(y)$

- The conditional entropy is $H(X|Y) = \sum_{i=1}^{4} p(Y = i) H(X|Y = i)$
- Similarly,
  $H(Y|X)$ = 13/8 bits and
  $H(X,Y)$ = 27/8 = 3.375 bits

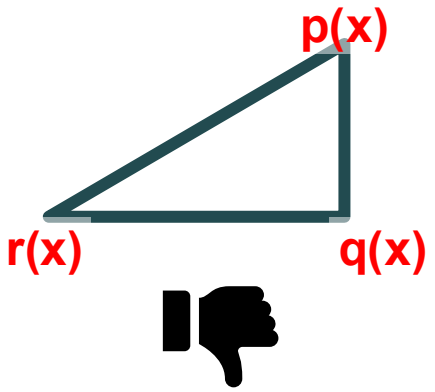
$$= \frac{1}{4} H\left(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}\right) + \frac{1}{4} H\left(\frac{1}{4}, \frac{1}{2}, \frac{1}{8}, \frac{1}{8}\right)$$
$$+ \frac{1}{4} H\left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right) + \frac{1}{4} H(1, 0, 0, 0)$$
$$= \frac{1}{4} \times \frac{7}{4} + \frac{1}{4} \times \frac{7}{4} + \frac{1}{4} \times 2 + \frac{1}{4} \times 0$$
$$= \frac{11}{8} \text{ bits.}$$

y = 1, y = 2, y = 3, y = 4

- N.B.: $H(Y|X) \neq H(X|Y)$
  - However
    $H(X) - H(X|Y) = H(Y) - H(Y|X)$

Information Distance

- The **entropy** of a RV is a measure of the **uncertainty** of the RV
  - It measures the amount of information required (on the average) to describe the RV
- The distance of information between two PMFs can be used to measure a difference
- Several formulas and definitions exist
- However, these are often not a true distance
  - a) they are not symmetric, i.e., $[q(x) \,\|\, p(x)] \neq [p(x) \,\|\, q(x)]$
  - b) they do not satisfy the triangle inequality, i.e., no $[r(x) \,\|\, p(x)] \leq [r(x) \,\|\, q(x)] + [q(x) \,\|\, p(x)]$
- Nonetheless, think of them as a "distance" between PMFs



Relative Entropy (Kullback–Leibler Distance)

- The **relative entropy (RE)** or **Kullback–Leibler distance (KLD)** or **information divergence** between two PMFs $p(x)$ and $q(x)$ is

Read as "from p to q"

$$D(p \parallel q) = D_{\mathrm{KL}}(p, q) = \sum_{x \in \mathcal{X}} p(x) \log \frac{p(x)}{q(x)} = \mathrm{E}_p \left\{ \log \frac{p(X)}{q(X)} \right\}$$

  - Convention: $0 \log(0/0) = 0$, $0 \log(0/q) = 0$, and $p \log(p/0) = \infty$
  - Thus, for $x \in \mathcal{X}$ such that $p(x) > 0$ and $q(x) = 0$, then $D(p\|q) = \infty$

- **Relative entropy is always nonnegative** and is 0 iff $p(x) = q(x)$
- It is also a **convex function** of $p(x)$
- The relative entropy ***D(p‖q)*** measures also the inefficiency of assuming that the PMF is q when the true PMF is p
  - If we knew the true PMF p of the RV, we could setup a questioning scheme requiring an average of $H(p)$ questions
  - If we designed the scheme assuming q, we would need $H(p) + D(p\|q)$ bits (on the average) to describe the RV

Cross-Entropy as Distance

- Closely related to KLD, **cross-entropy (CE)** is also a distance between two PMFs defined as

Note ||: not joint entropy

$$H(p \parallel q) = D_{\mathrm{CE}}(p,q) = -\sum_{x\in\mathcal{X}} p(x)\log q(x) = -\mathrm{E}_p\{\log q(X)\}$$

Note H: not KLD

$$= H(p) + D(p \parallel q)$$

$$H(p \parallel q) = -E_p\{\log q(X)\} = E_p\left\{\log\frac{1}{q(X)}\right\} + E_p\left\{\log\frac{p(X)}{p(X)}\right\}$$
$$= E_p\left\{\log\frac{1}{p(X)}\right\} + E_p\left\{\log\frac{1}{q(X)}\right\} + E_p\{\log p(X)\}$$
$$= H(p) + D(p \parallel q)$$

- CE measures the expected number of bits used to describe X with PMF $p(x)$ if we designed a scheme assuming PMF $q(x)$
- KLD, instead, measures the extra number of bits required for that
  - When KLD is zero, we are using the correct PMF to "schematize" the data or to "predict" future data
- CE is used in ML as *loss function* for classification tasks and in *operations research* to solve optimization problems

Example: KLD and CE

- Let $\mathcal{X} = \{0,1\}$ and consider two PMFs p and q on $\mathcal{X}$
- Let $p = \{1 - r, r\}$ and $q = \{1 - s, s\}$
- The KLDs and CEs are

$$D(p \parallel q) = (1-r)\log\frac{1-r}{1-s} + r\log\frac{r}{s} \qquad D(q \parallel p) = (1-s)\log\frac{1-s}{1-r} + s\log\frac{s}{r}$$

$$H(p \parallel q) = -(1-r)\log(1-s) - r\log s \qquad H(q \parallel p) = -(1-s)\log(1-r) - s\log r$$

- If $r = s$, then $D(p\|q) = D(q\|p) = H(q\|p) = H(p\|q) = 0$
- If $r = 1/2$, $s = 1/4$ we have

$$D(p \parallel q) = 0.2075 \qquad D(q \parallel p) = 0.1887$$

$$H(p \parallel q) = 1.2075 \qquad H(q \parallel p) = 1$$

$$D(p\|q) = \frac{1}{2}\log\frac{\frac{1}{2}}{\frac{3}{4}} + \frac{1}{2}\log\frac{\frac{1}{2}}{\frac{1}{4}} = 1 - \frac{1}{2}\log 3 = 0.2075 \text{ bit},$$

$$D(q\|p) = \frac{3}{4}\log\frac{\frac{3}{4}}{\frac{1}{2}} + \frac{1}{4}\log\frac{\frac{1}{4}}{\frac{1}{2}} = \frac{3}{4}\log 3 - 1 = 0.1887 \text{ bit}.$$

- Note that, as stated before, $D(p\|q) \neq D(q\|p)$ and $H(p\|q) \neq H(q\|p)$

Mutual Information

- **Mutual information (MI)** measures the amount of information that one RV contains (*knows*) about another RV
  - It is the reduction in the uncertainty of one RV due to the knowledge of the other RV
- Consider two RVs *X* and *Y* with a joint PMF $p(x,y)$ and marginal PMFs $p(x)$ and $p(y)$
- The **mutual information *I*(*X*;*Y*)** is the relative entropy between the joint PMF and the product PMF $p(x)p(y)$

$$I(X;Y) = \sum_{x\in\mathcal{X}}\sum_{y\in\mathcal{Y}} p(x,y)\log\frac{p(x,y)}{p(x)p(y)} = D\big(p(x,y)\parallel p(x)p(y)\big)$$

$$= \mathrm{E}_{p(x,y)}\left\{\log\frac{p(X,Y)}{p(X)p(Y)}\right\}$$

Relation between RE and MI

- Rewrite the definition of mutual information $I(X;Y)$ as

$$
\begin{aligned}
I(X;Y) &= \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p(x)p(y)} = \sum_{x,y} p(x,y) \log \frac{p(y)p(x|y)}{p(x)p(y)} \\
&= -\sum_{x,y} p(x,y) \log p(x) + \sum_{x,y} p(x,y) \log p(x|y) \\
&= -\sum_{x} p(x) \log p(x) - \left(-\sum_{x,y} p(x,y) \log p(x|y) \right) \\
&= H(X) - H(X|Y)
\end{aligned}
$$

- Thus, the mutual information $I(X;Y)$ is the reduction in the uncertainty of X due to the knowledge of Y

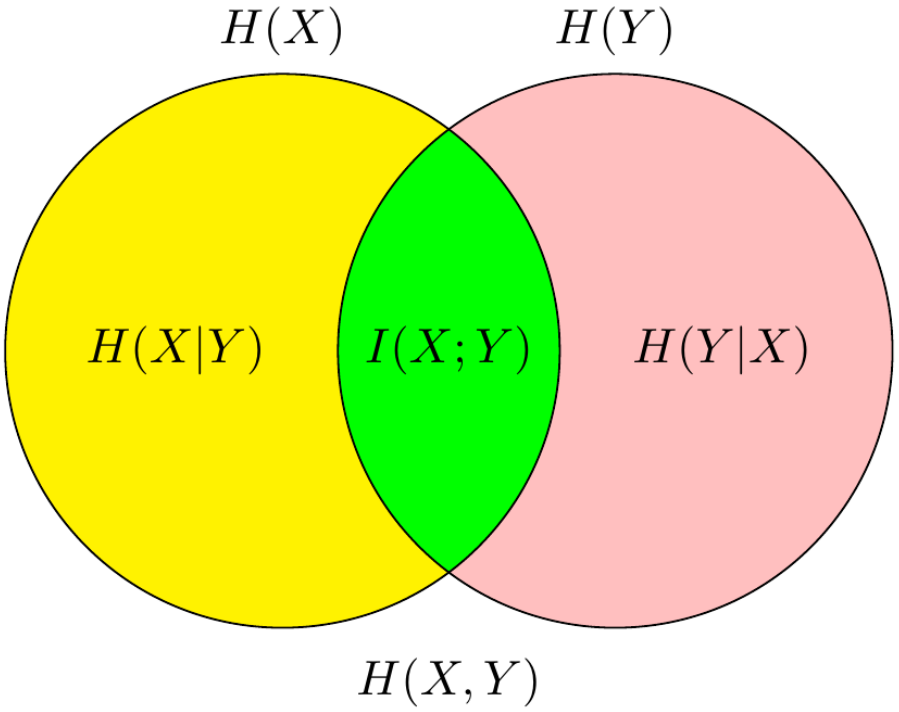
Relation between RE and MI

- Symmetrically, $I(X;Y) = H(Y) - H(Y|X)$
  - X says as much about Y as Y says about X
- Since $H(X,Y) = H(X) + H(Y|X)$, we have $I(X;Y) = H(X) + H(Y) - H(X,Y)$
- Also, $I(X;X) = H(X) - H(X|X) = H(X)$
  - The mutual information of a RV with itself is the entropy of the RV
- This explains why entropy is referred to as *self-information*

Mutual Information and Entropies

- It can be shown that graphical representations from set theory (**Venn diagrams**) correctly represents these equalities

$$I(X;Y) = H(X) - H(X|Y)$$
$$I(X;Y) = H(Y) - H(Y|X)$$
$$I(X;Y) = H(X) + H(Y) - H(X,Y)$$
$$I(X;Y) = I(Y;X)$$
$$I(X;X) = H(X)$$



- Note that the mutual information *I*(*X*;*Y*) corresponds to the intersection of the information in *X* with the information in *Y*

Example

- For the joint distribution used before

Y \ X	1	2	3	4
1	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$	$\frac{1}{32}$
2	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{32}$	$\frac{1}{32}$
3	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$
4	$\frac{1}{4}$	0	0	0

- The mutual information is

$$I(X;Y) = H(X) - H(X|Y) = 7/4 - 11/8$$
$$= H(Y) - H(Y|X) = 16/8 - 13/8$$
$$= 3/8 = 0.375 \text{ bits}$$

Chain Rule for Entropy

- **Theorem:** the entropy of a collection of RVs is the sum of the conditional entropies

$$H(X_1, X_2, \dots, X_n) = \sum_{i=1}^{n} H(X_i | X_{i-1}, \dots, X_1)$$

  - *Proof:* repeat the expansion rule for 2-RV entropies

$$H(X_1, X_2) = H(X_1) + H(X_2|X_1)$$

$$H(X_1, X_2, X_3) = H(X_1) + H(X_2, X_3|X_1) = H(X_1) + H(X_2|X_1) + H(X_3|X_2, X_1)$$

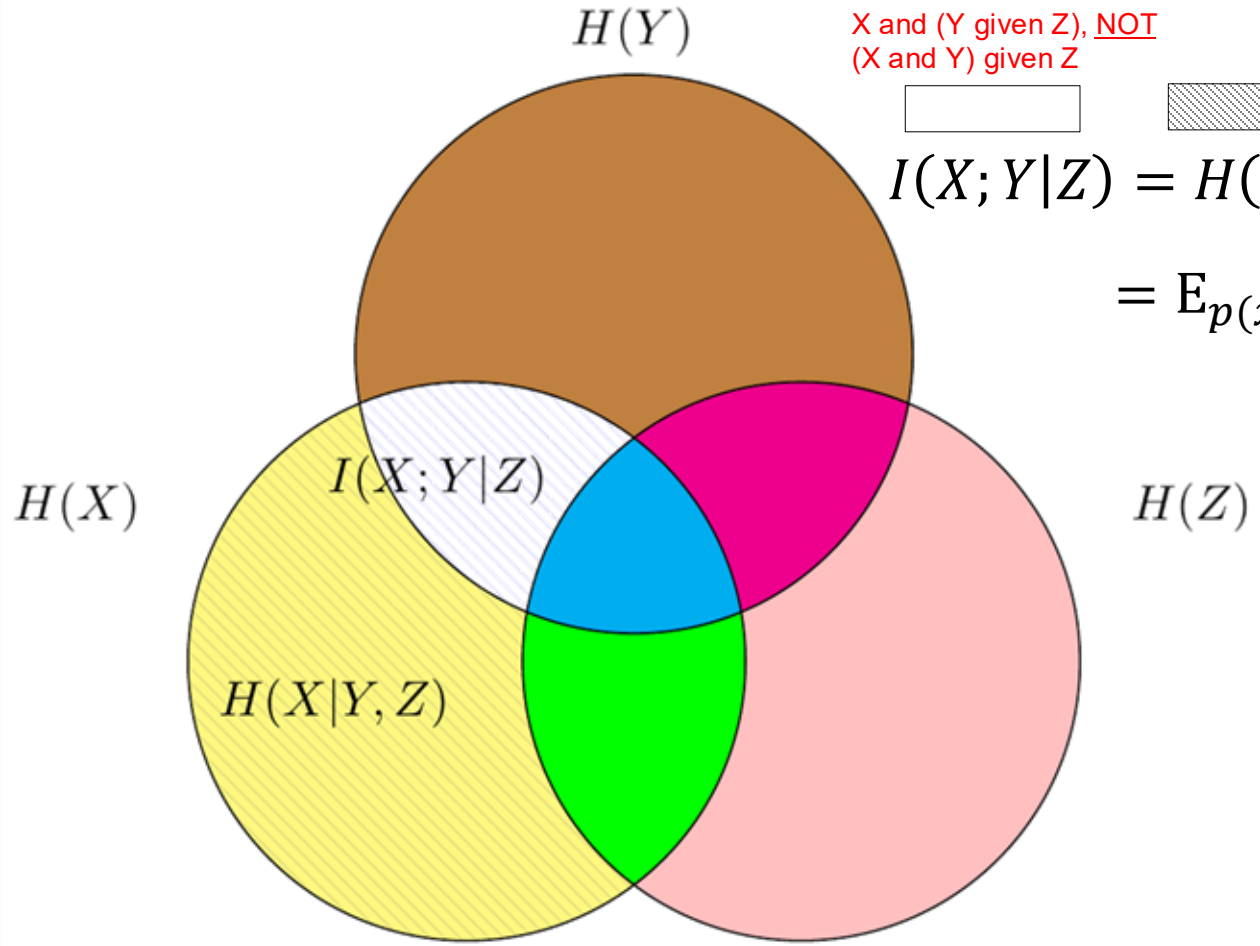
⋮

(X2 and X3) given X1, NOT X2 and (X3|X1)

$$H(X_1, X_2, \dots, X_n) = H(X_1) + H(X_2|X_1) + \cdots + H(X_n|X_{n-1}, \dots, X_1) = \sum_{i=1}^{n} H(X_i|X_{i-1}, \dots, X_1)$$

3 RVs: Conditional Mutual Information

- The **conditional mutual information** of RVs *X* and *Y* given *Z* is defined as the reduction in the uncertainty of *X* due to knowledge of *Y* when *Z* is given



X and (Y given Z), NOT
(X and Y) given Z

$$I(X;Y|Z) = H(X|Z) - H(X|Y,Z)$$

$$= \mathrm{E}_{p(x,y,z)}\left\{\log\frac{p(X,Y|Z)}{p(X|Z)p(Y|Z)}\right\}$$

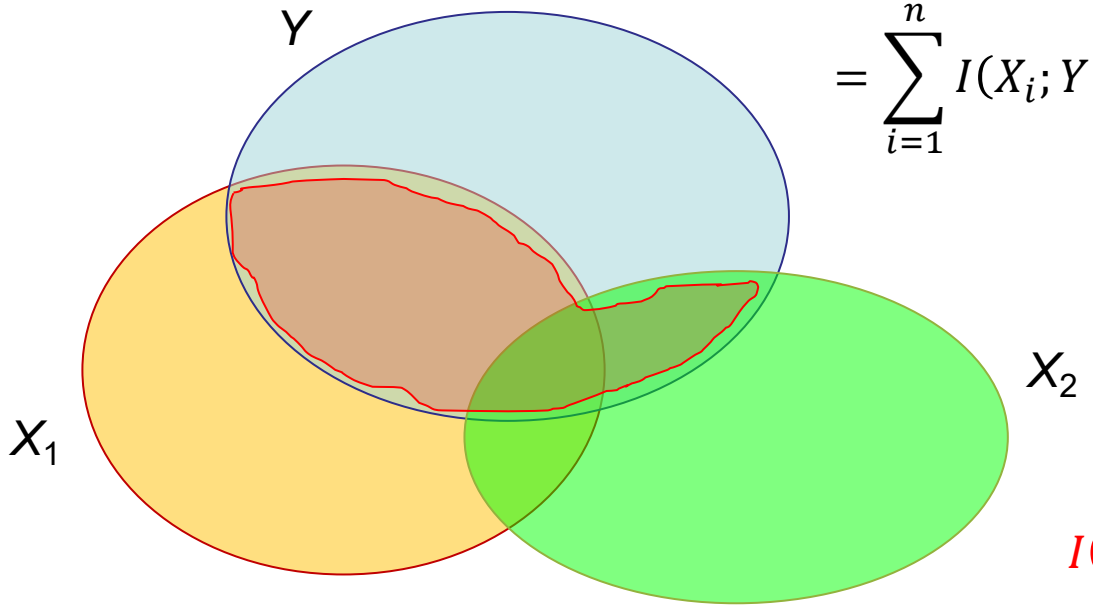
Chain Rule for Mutual Information

- **Theorem:** Chain rule for mutual information

$$I(X_1, X_2, \ldots, X_n; Y) = \sum_{i=1}^{n} I(X_i; Y | X_{i-1}, X_{i-2}, \ldots, X_1)$$

  - *Proof:*

$$\begin{aligned} I(X_1, X_2, \ldots, X_n; Y) &= H(X_1, X_2, \ldots, X_n) - H(X_1, X_2, \ldots, X_n | Y) \\ &= \sum_{i=1}^{n} H(X_i | X_{i-1}, \ldots, X_1) - \sum_{i=1}^{n} H(X_i | X_{i-1}, \ldots, X_1, Y) \\ &= \sum_{i=1}^{n} I(X_i; Y | X_1, X_2, \ldots, X_{i-1}) \end{aligned}$$



$$I(X_1, X_2; Y) = I(X_1; Y) + I(X_2; Y | X_1)$$

Conditional Relative Entropy

- For joint PMFs $p(x,y)$ and $q(x,y)$, the **conditional relative entropy** $D(p(y|x)\|q(y|x))$ is the average of the relative entropies between the conditional PMFs $p(y|x)$ and $q(y|x)$ averaged on the PMF $p(x)$

$$D\big(p(y|x) \,\|\, q(y|x)\big) = \sum_x p(x) \sum_y p(y|x) \log \frac{p(y|x)}{q(y|x)}$$

$$= \mathrm{E}_{p(x,y)} \left\{ \log \frac{p(Y|X)}{q(Y|X)} \right\}$$

- The notation is not explicit since it omits the PMF $p(x)$ of the conditioning RV
  - However, it is normally understood from the context

Chain Rule for Relative Entropy

- **Theorem:** the relative entropy between two joint PMFs on a pair of RVs can be expanded as the sum of a relative entropy and a conditional relative entropy

$$D\big(p(x,y) \parallel q(x,y)\big) = D\big(p(x) \parallel q(x)\big) + D\big(p(y|x) \parallel q(y|x)\big)$$

  - Proof:

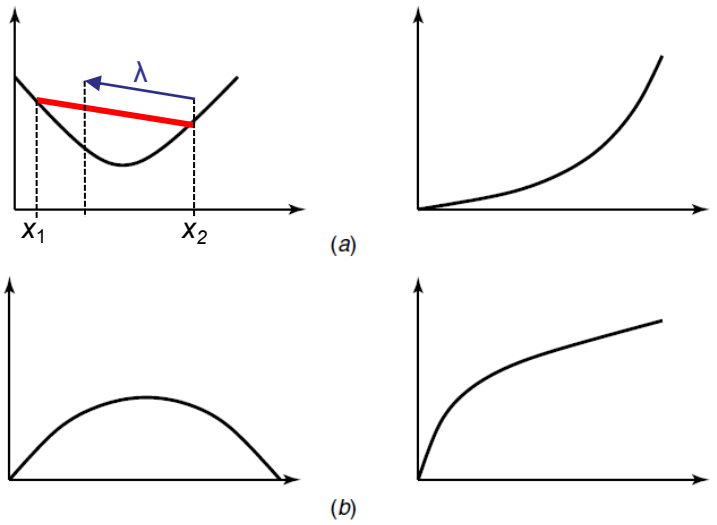
$$\begin{aligned} D\big(p(x,y) \parallel q(x,y)\big) &= \sum_x \sum_y p(x,y) \log \frac{p(x,y)}{q(x,y)} = \sum_x \sum_y p(x,y) \log \frac{p(x)p(y|x)}{q(x)q(y|x)} \\ &= \sum_x \sum_y p(x,y) \log \frac{p(x)}{q(x)} + \sum_x \sum_y p(x,y) \log \frac{p(y|x)}{q(y|x)} \\ &= D\big(p(x) \parallel q(x)\big) + D\big(p(y|x) \parallel q(y|x)\big) \end{aligned}$$

Concavity and Convexity

- A function $f(x)$ is **convex** over an interval (a,b) if for every $x_1, x_2 \in (a,b)$ and $0 \leq \lambda \leq 1$

$$f(\lambda x_1 + (1-\lambda)x_2) \leq \lambda f(x_1) + (1-\lambda) f(x_2)$$

  - A function is convex if it always lies below any chord
- f is **strictly convex** if equality holds only if $\lambda = 0$ or $\lambda = 1$
- f is **concave** if $-f$ is convex
  - A function is concave if it always lies above any chord
- Convex: x^2, $|x|$, e^x, $x \log(x)$ for $x \geq 0$, etc.
- Concave: $\log(x)$ and $\sqrt{x}$ for $x \geq 0$
- Note that linear functions $ax + b$ are both convex and concave
- Many properties of entropy, mutual information, etc., underlie to convexity



Convexity Theorem

- **Theorem:** If the function *f* has a second derivative that is nonnegative (positive) over an interval, the function is convex (strictly convex) over that interval

  - *Proof:* Taylor expansion around x_0 with x^* between x_0 and x

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x^*)}{2}(x - x_0)^2$$ (with Lagrange remainder)

  By hypothesis, $f''(x^*) \geq 0$: the last term is nonnegative for all x.
  Let $x_0 = \lambda x_1 + (1 - \lambda)\, x_2$ and take $x = x_1$

$$f(x_1) \geq f(x_0) + f'(x_0)\big((1 - \lambda)(x_1 - x_2)\big)$$

  Similarly take $x = x_2$ to obtain $f(x_2) \geq f(x_0) + f'(x_0)(\lambda(x_2 - x_1))$
  Multiplying by λ and by $1 - \lambda$ and adding, we obtain the definition

$$\lambda f(x_1) + (1 - \lambda) f(x_2) \geq f(\lambda x_1 + (1 - \lambda) x_2)$$

- The proof for strict convexity is similar

Convexity Examples

- Strict convexity of x^2, e^x, and $x \log(x)$ for $x \geq 0$

$$\frac{d^2}{dx^2} x^2 > 0$$
$$\frac{d}{dx} 2x > 0$$
$$2 > 0$$
$$x \in \mathbb{R} \wedge \mathbb{R}$$

$$\frac{d^2}{dx^2} e^x > 0$$
$$\frac{d}{dx} e^x > 0$$
$$e^x > 0$$
$$x \in \mathbb{R} \wedge \mathbb{R}$$

$$\frac{d^2}{dx^2} x \ln x > 0$$
$$\frac{d}{dx}\left(x \frac{1}{x} + \ln x \right) > 0$$
$$\frac{1}{x} > 0$$
$$x \in \mathbb{R}^+ \wedge \mathbb{R}^+$$

- Strict concavity of $\log(x)$ and $\sqrt{x}$ for $x \geq 0$

$$\frac{d^2}{dx^2} \ln x < 0$$
$$\frac{d}{dx} \frac{1}{x} < 0$$
$$-\frac{1}{x^2} < 0$$
$$x^2 > 0$$
$$x \in \mathbb{R}^+ \wedge \mathbb{R}$$

$$\frac{d^2}{dx^2} \sqrt{x} < 0$$
$$\frac{d}{dx} \frac{1}{2\sqrt{x}} < 0$$
$$-\frac{1}{4\sqrt{x^3}} < 0$$
$$x^3 > 0$$
$$x \in \mathbb{R}^+ \wedge \mathbb{R}^+$$

Jensen's Inequality

- **Theorem:** If $f()$ is a convex function and X is a RV

$$\mathrm{E}\{f(X)\} \geq f(\mathrm{E}\{X\})$$

If $f()$ is strictly convex, the equality implies that $X = \mathrm{E}\{X\}$ (constant)

- *Proof:* For a 2-mass PMF, the inequality is true for definition of convexity

$$p_1 f(x_1) + p_2 f(x_2) \geq f(p_1 x_1 + p_2 x_2)$$

Suppose that the theorem is true for PMFs with $k-1$ mass points. Then writing $p'_i = p_i/(1-p_k)$, $i = 1,2,\ldots,k-1$, by using induction and with the definition of convexity, we have

$$\sum_{i=1}^{k} p_i f(x_i) = p_k f(x_k) + (1-p_k) \sum_{i=1}^{k-1} p'_i f(x_i) \geq p_k f(x_k) + (1-p_k) f\left(\sum_{i=1}^{k-1} p'_i x_i \right)$$

true for k-1 masses; mass 1; mass 2

$$\geq f\left(p_k x_k + (1-p_k) \sum_{i=1}^{k-1} p'_i x_i \right) = f\left(\sum_{i=1}^{k} p_i x_i \right)$$

true for 2 masses

- Proof can be extended to continuous RV using continuity

Information Inequality

- **Theorem:** Let $p(x)$, $q(x)$, $x \in \mathcal{X}$, be two PMFs. Then

$$D(p \parallel q) \geq 0$$

with equality iff $p(x) = q(x)$ for all x

- *Proof:* Let $\mathcal{A} = \{x : p(x) > 0\}$ be the support set of $p(x)$

$$-D(p \parallel q) = -\sum_{x \in \mathcal{A}} p(x) \log \frac{p(x)}{q(x)} = \sum_{x \in \mathcal{A}} p(x) \log \frac{q(x)}{p(x)} \leq \log \sum_{x \in \mathcal{A}} p(x) \frac{q(x)}{p(x)}$$

Jensen's inequality (concave) where $f(x')$ is $\log(x')$ and $x' = q(x)/p(x)$

$$= \log \sum_{x \in \mathcal{A}} q(x) \leq \log \sum_{x \in \mathcal{X}} q(x) = \log 1 = 0$$

Since log() is strictly concave, there is equality iff $q(x)/p(x)$ is constant, i.e., $q(x) = cp(x)$. There is another equality if $c = 1$.
Hence, we have $D(p\|q) = 0$ iff $p(x) = q(x)$ for all x

Corollaries

- (*Nonnegativity of mutual information*) For any two RVs X and Y

$$I(X;Y) \geq 0$$

with equality iff X and Y are independent

  - *Proof:* $I(X;Y) = D(p(x,y)\|p(x)p(y)) \geq 0$, with eq. iff $p(x,y) = p(x)p(y)$

- $$D(p(y|x) \parallel q(y|x)) \geq 0$$

with equality iff $p(y|x) = q(y|x)$ for all y and x such that $p(x) > 0$

- $$I(X;Y|Z) \geq 0$$

with equality iff X and Y are conditionally independent given Z

Maximum Entropy of Uniform Distribution

cardinality

- **Theorem:** $H(X) \leq \log|\mathcal{X}|$, where $|\mathcal{X}|$ denotes the number of elements in the range of X, with equality iff X has a uniform distribution over $\mathcal{X}$
  - *Proof:* Let $u(x) = 1/|\mathcal{X}|$ be the uniform PMF over $\mathcal{X}$, and let $p(x)$ be the probability mass function for X

$$D(p \parallel u) = \sum_{x \in \mathcal{X}} p(x) \log \frac{p(x)}{u(x)} = \log|\mathcal{X}| - H(X)$$

Hence by the nonnegativity of relative entropy

$$0 \leq D(p \parallel u) = \log|\mathcal{X}| - H(X)$$

- The uniform PMF over the range $\mathcal{X}$ is the maximum entropy distribution over this range
- Any RV with this range has an entropy no greater than $\log|\mathcal{X}|$

Conditioning Reduces Entropy

- **Theorem:** aka "**Information can't hurt**"

$$H(X|Y) \leq H(X)$$

with equality iff X and Y are independent

  - *Proof:* $0 \leq I(X;Y) = H(X) - H(X|Y)$

- Knowing another random variable Y can only reduce the uncertainty in X, <u>but this is true only on the average</u>
- $H(X|Y = y)$ may be greater than or less than or equal to $H(X)$, but on the average $H(X|Y) = \Sigma_y p(y)\, H(X|Y = y) \leq H(X)$
  - E.g., in court, new evidence might increase uncertainty, but on the average evidence decreases uncertainty



Example: Coin Toss

- Consider tossing a coin extracted from a bag of coins of unknown fairness
  - X is the side
  - Y is the fairness
- Let (X,Y) have the joint PMF

Y \ X	HEADS 1	TAILS 2
RIGGED 1	0	$\frac{3}{4}$
FAIR 2	$\frac{1}{8}$	$\frac{1}{8}$



- Unconditional uncertainty is $H(X) = H(1/8,7/8) = 0.544$ bits
- Conditional uncertainties are $H(X|Y = 1) = 0$ and $H(X|Y = 2) = 1$
- On average, $H(X|Y) = (3/4)\ \mathrm{H}(X|Y = 1) + (1/4)\ \mathrm{H}(X|Y = 2) = 0.25$ bit
- Thus, uncertainty in X is
  - increased if $Y = 2$ is observed: fair coin
  - decreased if $Y = 1$ is observed: rigged coin
- Uncertainty decreases (from 0.544 to 0.25) on average

Independence Bound on Entropy

- **Theorem:** Let $X_1,X_2,\ldots,X_n$ be drawn according to $p(x_1,x_2,\ldots,x_n)$

$$H(X_1, X_2, \ldots, X_n) \leq \sum_{i=1}^{n} H(X_i)$$

with equality iff the X_i are independent

  - *Proof:* by the chain rule for entropies

$$H(X_1, X_2, \ldots, X_n) = \sum_{i=1}^{n} H(X_i|X_{i-1}, \ldots, X_1) \leq \sum_{i=1}^{n} H(X_i)$$

Information cannot hurt

where the inequality follows from "Information can't hurt"

- Equality holds iff X_i is independent of $X_i-1,\ldots,X_1$ for all i (i.e., iff the X_i's are independent)

Log Sum Inequality

- **Theorem:** For nonnegative numbers $a_1,a_2,\ldots,a_n$ and $b_1,b_2,\ldots,b_n$,

$$\sum_{i=1}^{n} a_i \log \frac{a_i}{b_i} \geq \left(\sum_{i=1}^{n} a_i\right) \log \frac{\sum_{i=1}^{n} a_i}{\sum_{i=1}^{n} b_i}$$

with equality iff $a_i/b_i = const$

- *Proof:* We use $0 \log 0 = 0$, $a \log(a/0) = \infty$ if $a > 0$ and $0 \log(0/0) = 0$

Assume (without loss of generality) that $a_i > 0$ and $b_i > 0$

The function $f(t) = t \log(t)$ is strictly convex for all positive t. By Jensen's inequality

$$\sum_i \alpha_i f(t_i) \geq f\left(\sum_i \alpha_i t_i\right)$$

for $\alpha_i \geq 0$, $\Sigma_i \alpha_i = 1$

- Let $\alpha_i = b_i / \Sigma_{j=1}^{n} b_j$ and $t_i = a_i/b_i$

Log Sum Inequality (proof cont'd)

$$\sum_i a_i \log \frac{a_i}{b_i} = \sum_i b_i \frac{a_i}{b_i} \log \frac{a_i}{b_i} = \sum_i b_i t_i \log t_i = \sum_i \left(\alpha_i \sum_j b_j \right) t_i \log t_i$$

$$= \left(\sum_j b_j \right) \sum_i \alpha_i t_i \log t_i = \left(\sum_j b_j \right) \sum_i \alpha_i f(t_i) \geq \left(\sum_j b_j \right) f\left(\sum_i \alpha_i t_i \right)$$

$$= \left(\sum_j b_j \right) \left(\sum_i \alpha_i t_i \right) \log \left(\sum_i \alpha_i t_i \right) = \left(\sum_j b_j \right) \left(\sum_i \frac{b_i}{\sum_j b_j} \frac{a_i}{b_i} \right) \log \left(\sum_i \frac{b_i}{\sum_j b_j} \frac{a_i}{b_i} \right)$$

$$= \sum_i \left(\left(\sum_j b_j \right) \frac{b_i}{\sum_j b_j} \frac{a_i}{b_i} \right) \log \left(\sum_i \frac{b_i}{\sum_j b_j} \frac{a_i}{b_i} \right) = \left(\sum_i a_i \right) \log \left(\frac{\sum_i a_i}{\sum_j b_j} \right)$$

Convexity of Relative Entropy

- **Theorem:** $D(p\|q)$ is **convex** in the pair (p,q); that is, if (p_1,q_1) and (p_2,q_2) are two pairs of PMFs, then for all $0 \le \lambda \le 1$

$$D(\lambda p_1 + (1-\lambda)p_2 \parallel \lambda q_1 + (1-\lambda)q_2) \le \lambda D(p_1 \parallel q_1) + (1-\lambda)D(p_2 \parallel q_2)$$

  - *Proof:* apply the log sum inequality to one right-hand side addend

$$\underset{\Sigma_{i=1}^{2} a_i}{\left(\lambda p_1(x) + (1-\lambda)p_2(x)\right)} \log \frac{\overset{\Sigma_{i=1}^{2} a_i}{\lambda p_1(x) + (1-\lambda)p_2(x)}}{\underset{\Sigma_{i=1}^{2} b_i}{\lambda q_1(x) + (1-\lambda)q_2(x)}}$$

$$\le \overset{a_1}{\lambda p_1(x)} \log \frac{\overset{a_1}{\lambda p_1(x)}}{\underset{b_1}{\lambda q_1(x)}} + \overset{a_2}{(1-\lambda)p_2(x)} \log \frac{\overset{a_2}{(1-\lambda)p_2(x)}}{\underset{b_2}{(1-\lambda)q_2(x)}}$$

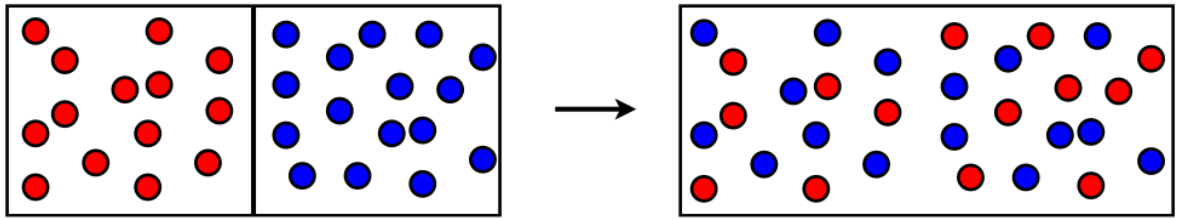
  - Summing this over all x, we obtain the desired property

Concavity of Entropy

- **Theorem:** $H(p)$ is a concave function of p
  - Proof: $H(p) = \log|\mathcal{X}| - D(p\|u)$
    where u is the uniform distribution on $|\mathcal{X}|$ outcomes. The concavity of H then follows directly from the convexity of KLD
- This means that

$$H(\lambda p + (1-\lambda)q) \geq \lambda H(p) + (1-\lambda)H(q)$$

- Equivalence in statistical mechanics: due to the concavity of entropy, mixing two gases of equal entropy results in a gas with higher entropy
  - Spatial uncertainty on which gas occupies a minuscule volume



Mutual Information Concavity

- **Theorem:** Let $(X,Y) \sim p(x,y) = p(x)\,p(y|x)$. The mutual information $I(X;Y)$ is a concave function of $p(x)$ for fixed $p(y|x)$ and a convex function of $p(y|x)$ for fixed $p(x)$

  - *Proof:* for the 1st part, expand the mutual information

Concave in p(x)

Linear in p(x)

$$I(X;Y) = H(Y) - H(Y|X) = H(Y) - \sum_x p(x)H(Y|X=x)$$

p(y) = p(x) p(y|x)
Linear in p(x)

If $p(y|x)$ is fixed, then $p(y)$ is a linear function of $p(x)$. Hence $H(Y)$, a concave function of $p(y)$, is a concave function of $p(x)$. The 2nd term is a linear function of $p(x)$. Hence, the difference is a concave function of $p(x)$

  - For the 2nd part, fix $p(x)$ and consider two different conditional PMFs $p_1(y|x)$ and $p_2(y|x)$, their joint PMFs $p_1(x,y)=p(x)p_1(y|x)$ and $p_2(x,y)=p(x)p_2(y|x)$, and their marginal PMFs $p(x)$, $p_1(y)$ and $p(x)$, $p_2(y)$. Consider a conditional PMF to be a mixture of $p_1(y|x)$ and $p_2(y|x)$, where $0 \le \lambda \le 1$

$$p_\lambda(y|x) = \lambda p_1(y|x) + (1-\lambda)p_2(y|x)$$

Mutual Information Concavity (proof cont'd)

- The joint PMF is also a mixture of the joint PMFs,

$$p_\lambda(x,y) = \lambda p_1(x,y) + (1-\lambda)p_2(x,y)$$

and the distribution of *Y* is also a mixture,

$$p_\lambda(y) = \lambda p_1(y) + (1-\lambda)p_2(y)$$

- Hence if we let $q_\lambda(x,y) = p(x)\, p_\lambda(y)$ be the product of the marginal distributions, we have

$$q_\lambda(x,y) = \lambda q_1(x,y) + (1-\lambda)q_2(x,y)$$

- Since the mutual information is the relative entropy between the joint PMF and the product of the marginals,

$$I(X;Y) = D\big(p_\lambda(x,y) \parallel q_\lambda(x,y)\big)$$

and relative entropy $D(p\|q)$ is a convex function of (p,q), it follows that the mutual information is a convex function of the conditional PMF

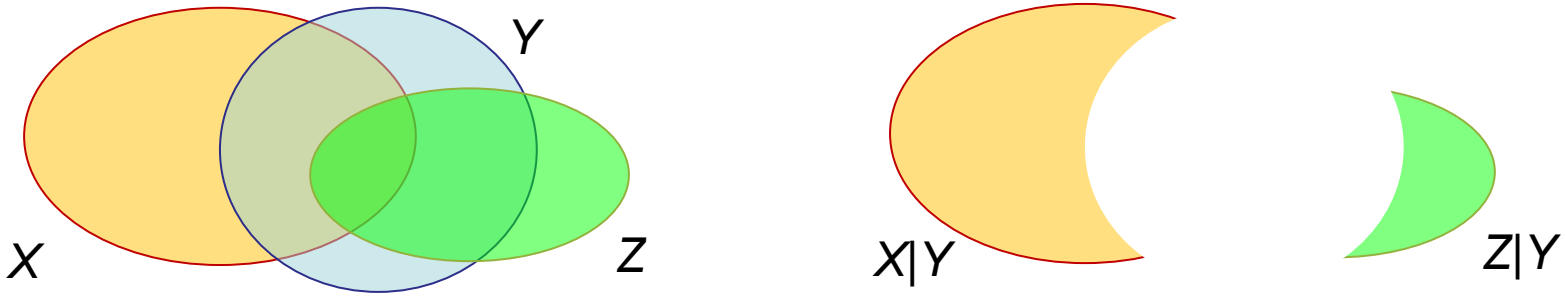
Markov Chains

- RVs X,Y,Z form a **Markov chain** in the order $X \rightarrow Y \rightarrow Z$ if the conditional PMF of Z depends only on Y and is conditionally independent of X, such that the joint PMF satisfies

$$p(x,y,z) = p(x)p(y|x)p(z|y)$$

- $X \rightarrow Y \rightarrow Z$ iff X and Z are conditionally independent given Y (Markovity implies conditional independence)

$$p(x,z|y) = \frac{p(x,y,z)}{p(y)} = \frac{p(x,y)p(z|y)}{p(y)} = \frac{p(y)p(x|y)p(z|y)}{p(y)}$$
$$= p(x|y)p(z|y)$$



Example: Markov Chain

- Consider the Bernoulli RV X with $x \in \{0,1\}$ and PMF $p(x)$

$$p(0) = 0.5, \qquad p(1) = 0.5$$

- Another Bernoulli RV Y depends on X with $q(y|x)$ as

$$q(0|0) = 0.7, \qquad q(1|0) = 0.3$$
$$q(0|1) = 0.4, \qquad q(1|1) = 0.6$$

- Finally, a third RV Z depends on Y, but not on X, with $r(z|y)$ as

$$r(0|0) = 0.8, \qquad r(1|0) = 0.2$$
$$r(0|1) = 0.3, \qquad r(1|1) = 0.7$$

- So, r=1 with no specific dependence on the value assumed by x
  - X = sunny day (0) or rainy day (1) in SF
  - Y = ride a bike (0) or drive a car (1)
  - Z = take a friend (0) or the partner (1)

x	y	z	$p(x,y,z)$
0	0	0	0.5×0.7×0.8=0.28
0	0	1	0.5×0.7×0.2=0.07
0	1	0	0.5×0.3×0.3=0.045
0	1	1	0.5×0.3×0.7=0.105
1	0	0	0.5×0.4×0.8=0.16
1	0	1	0.5×0.4×0.2=0.04
1	1	0	0.5×0.6×0.3=0.09
1	1	1	0.5×0.6×0.7=0.21

Markov Chain Properties

- $X \rightarrow Y \rightarrow Z$ implies that $Z \rightarrow Y \rightarrow X$, written as $X \leftrightarrow Y \leftrightarrow Z$
  - *Proof*:

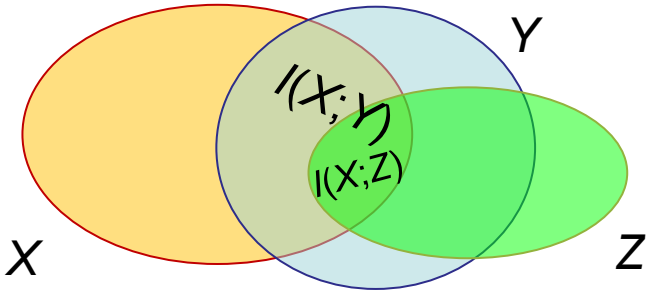
$$p(x,y,z) = p(x)p(y|x)p(z|x,y) = p(x)p(y|x)p(z|y)$$
$$= p(x)\frac{p(x|y)p(y)}{p(x)}\frac{p(y|z)p(z)}{p(y)} = p(z)p(y|z)p(x|y)$$

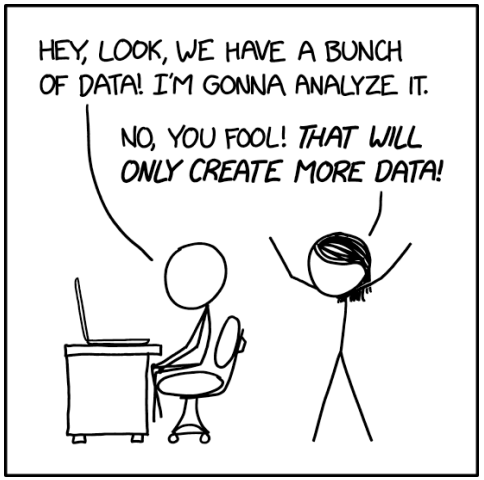
- If $Z = f(Y)$, i.e., a deterministic RV transformation, then $X \rightarrow Y \rightarrow Z$
  - *Proof*:

$$p(z|x,y) = \Pr\{Z = z|X = x, Y = y\} = \Pr\{Z = f(y)|X = x, Y = y\}$$
$$= \Pr\{Z = f(y)|Y = y\} = p(z|y)$$

Data Processing Inequality

- For the folks
  - No processing of *Y*, deterministic or random, can increase the information that *Y* contains about *X*
  - No clever manipulation of the data can improve the inferences that can be made from the data





- **Theorem:** If $X \rightarrow Y \rightarrow Z$, then $I(X;Y) \geq I(X;Z)$
  - *Proof:* By the chain rule on mutual information

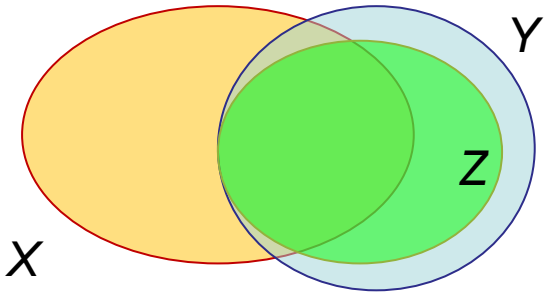
$$I(X;Y,Z) = I(X;Z) + I(X;Y|Z) = I(X;Y) + I(X;Z|Y)$$

X and *Z* are conditionally independent given *Y*, so $I(X;Z|Y) = 0$. Since $I(X;Y|Z) \geq 0$, we have $I(X;Y) \geq I(X;Z)$

  - Equality iff $I(X;Y|Z) = 0$ (i.e., $X \rightarrow Z \rightarrow Y$). Similarly, $I(Y;Z) \geq I(X;Z)$

Corollaries of DPI

- If $Z = g(Y)$, we have $I(X;Y) \geq I(X;g(Y))$
  - *Proof:* $X \rightarrow Y \rightarrow g(Y)$ forms a Markov chain
  - Functions of the data Y cannot increase the information about X



- If $X \rightarrow Y \rightarrow Z$, then $I(X;Y|Z) \leq I(X;Y)$
  - *Proof:* We note in data proc. inequal. that $I(X;Z|Y) = 0$, by Markovity, and $I(X;Z) \geq 0$. Thus, $I(X;Y|Z) \leq I(X;Y)$
  - The dependence of X and Y is decreased (or remains unchanged) by the observation of a "downstream" random variable Z
  - It is also possible that $I(X;Y|Z) > I(X;Y)$ when X,Y,Z are not a Markov chain

0+0=0 0+1=1 1+1=2 1+0=1

- E.g., let X and Y be independent fair binary RVs, and let $Z = X + Y$. Then $I(X;Y) = 0$, but

H(X|Z=0)=0 H(X|Z=1)=1 H(X|Z=2)=0

$I(X;Y|Z) = H(X|Z) - H(X|Y,Z) = H(X|Z) = P(Z=1)\, H(X|Z=1) = 1/2$ bit

Statistic

- Consider a family of PMFs $\{f_\theta(x)\}$ indexed by the random parameter θ, and let X be samples from a PMF in this family
- Let $T(X)$ be any **statistic**, i.e., a function of the samples $X_1, X_2, \ldots$
  - Sample mean $T_1(X) = \bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$
  - Sample variance $T_2(X) = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X}_n)^2$
  - Maximum sample $T_3(X) = \max\{X_1, \ldots, X_n\}$
  - Weird function independent of samples $T_4(X) = 5$
- Then $\theta \rightarrow X \rightarrow T(X)$, and by the data-processing inequality,

$$I(\theta;T(X)) \leq I(\theta;X)$$

  for any distribution on θ
- However, <u>if equality holds, no information is lost</u>

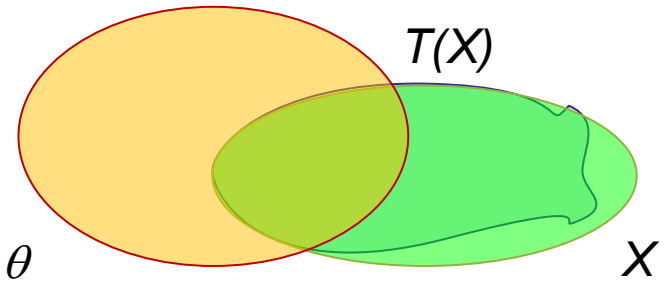
Sufficient Statistics

- A function $T(X)$ is said to be a **sufficient statistic** relative to the family $\{f_\theta(x)\}$ if X is independent of θ given $T(X)$ for any distribution on θ: i.e., $\theta \rightarrow T(X) \rightarrow X$ forms a Markov chain
  - $T(X)$ contains all the information in X about θ

- This is the same as the condition for equality in the data-processing inequality for all distributions on θ

$$I(\theta; X) = I(\theta; T(X))$$

- Hence sufficient statistics preserve mutual information and vice-versa



Factorization Theorem

- The proof of sufficiency can be tough, especially for continuous RVs, but there is a theorem

- **Theorem (Neyman-Fisher):** a statistic $T(X_1,X_2,\ldots,X_n)$ is sufficient iff the joint distribution can be factored as

$$f(x_1,\ldots,x_n|\theta) = u(x_1,\ldots,x_n)v(T(x_1,\ldots,x_n),\theta)$$

  where $u(\cdot)$ and $v(\cdot)$ are nonnegative functions

- This means that the joint distribution can be written as the product of
  - a function $u(\cdot)$ that can depend on the samples $x_1,x_2,\ldots,x_n$ but must not depend on the parameter θ
  - a function $v(\cdot)$ that can depend on the parameter θ, and can depend on the samples only via the statistic $T(x_1,x_2,\ldots,x_n)$

Example of Sufficient Statistics for Bernoulli PMF

- Let $X_1, X_2, \ldots, X_n$, $X_i \in \{0,1\}$, be an independent and identically distributed (i.i.d.) sequence of coin tosses of a coin with unknown parameter $\theta = \Pr(X_i = 1)$
- Given n, the number of 1's is a sufficient statistic for θ. Here $T(X_1, X_2, \ldots, X_n) = \sum_{i=1}^{n} X_i$
- Given T, all sequences having that many 1's are equally likely and independent of the parameter θ

$$\Pr\{(X_1, X_2, \ldots, X_n) = (x_1, x_2, \ldots, x_n) | \sum_{i=1}^{n} X_i = k\} = \begin{cases} \dfrac{1}{\binom{n}{k}} & \sum_{i=1}^{n} x_i = k \\ 0 & \text{otherwise} \end{cases}$$

- Thus, $\theta \rightarrow \sum_{i=1}^{n} X_i \rightarrow (X_1, X_2, \ldots, X_n)$ forms a Markov chain, and T is a sufficient statistic for θ

Example of Sufficient Statistics for Normal PDF

- If X is normally distributed with mean θ and variance 1; that is, if

$$f_\theta(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-\theta)^2}{2}} = \mathcal{N}(\theta, 1)$$

and $X_1, X_2, \ldots, X_n$ are drawn independently according to this distribution, a sufficient statistic for θ is the sample mean

$$T(X_1, \ldots, X_n) = \bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \qquad f(\bar{x}_n|\theta, n) = \sqrt{\frac{n}{2\pi}} e^{-n\frac{(\bar{x}_n-\theta)^2}{2}} = \mathcal{N}\left(\theta, \frac{1}{n}\right)$$

- It can be verified that the conditional distribution of $X_1, X_2, \ldots, X_n$, conditioned on $\bar{X}_n$ and n, does not depend on θ
- Simply, the factorization theorem holds, since

$$\begin{aligned} f(x_1, \ldots, x_n|\theta) &= (2\pi)^{-\frac{n}{2}} e^{-\frac{1}{2}\sum_{i=1}^{n}(x_i-\theta)^2} = (2\pi)^{-\frac{n}{2}} e^{-\frac{1}{2}\sum_{i=1}^{n} x_i^2 + \theta\sum_{i=1}^{n} x_i - \frac{n\theta^2}{2}} \\ &= (2\pi)^{-\frac{n}{2}} e^{-\frac{1}{2}\sum_{i=1}^{n} x_i^2} e^{-\frac{n\theta^2}{2} + \theta\sum_{i=1}^{n} x_i} = u(x_1, \ldots, x_n) e^{-\frac{n\theta^2}{2} + \theta n T(x_1, \ldots, x_n)} \\ &= u(x_1, \ldots, x_n) v(T(x_1, \ldots, x_n), \theta) \end{aligned}$$

Example of Sufficient Statistics for Uniform PDF

- For a uniform pdf, $f_\theta = \mathcal{U}(\theta, \theta+1)$, a sufficient statistic for θ is

$$T(X_1, X_2, \ldots, X_n) = \left(T_1(X), T_2(X)\right) = \left(\max\{X_1, X_2, \ldots, X_n\}, \min\{X_1, X_2, \ldots, X_n\}\right)$$

- The proof is slightly more complicated, but the distribution of the data is independent of the parameter given the statistic *T*

- Use the factorization theorem instead

$$f(x_1, \ldots, x_n|\theta) = \begin{cases} 1 & \theta \le x_i < \theta + 1, \forall i \\ 0 & \text{else} \end{cases}$$

$$= (1 + 0x_1 + \cdots + 0x_n) \begin{cases} 1 & (\theta \le \min x_i) \wedge (\max x_i < \theta + 1) \\ 0 & \text{else} \end{cases}$$

$$= u(x_1, \ldots, x_n) v(T(x_1, \ldots, x_n), \theta)$$

Minimal Sufficient Statistic

- A statistic $T(X)$ is a **minimal sufficient statistic** relative to $\{f_\theta(x)\}$ if it is a function $T(X) = \psi(U(X))$ of every other sufficient statistic $U(X)$, with $\psi()$ measurable

- In terms of the data-processing inequality, this implies that

$$\theta \longrightarrow T(X) \longrightarrow U(X) \longrightarrow X$$

- A minimal sufficient statistic maximally compresses the information about θ in the sample
  - In previous examples, the sufficient statistics are also minimal

- Other sufficient statistics may contain additional irrelevant information

Example: Nonminimal Sufficient Statistic

- For $\mathcal{N}(\theta,*)$, the pair of functions giving the mean of all odd samples and the mean of all even samples is a sufficient statistic, but <u>not</u> a minimal sufficient statistic

Sufficient, but not minimal

$$T'(X_1,\dots,X_{2n}) = (T'_{\mathrm{O}}, T'_{\mathrm{E}}) = \left(\frac{1}{n}\sum_{i=1}^{n} X_{2i-1}, \frac{1}{n}\sum_{i=1}^{n} X_{2i}\right) \neq \big(\varphi(T), \psi(T)\big)$$

$$T(X_1,\dots,X_{2n}) = \frac{1}{2n}\sum_{i=1}^{2n} X_i = \frac{T'_{\mathrm{O}} + T'_{\mathrm{E}}}{2}$$

Sufficient and minimal

- Indeed, you cannot write $T'_{\mathrm{O}}()$ or $T'_{\mathrm{E}}()$ as function of $T()$

Fano's Inequality

- We know a RV *Y* and we wish to guess a correlated RV *X*
- Fano's inequality relates the probability of error in guessing the RV *X* to its conditional entropy $H(X|Y)$
- It can be shown that **the conditional entropy of a RV X given another RV Y is zero iff *X* is a function of *Y* only**
- We can estimate *X* from *Y* with 0 probability of error **iff** $H(X|Y)=0$
- We expect to be able to estimate *X* with a low probability of error only if the conditional entropy $H(X|Y)$ is small
  - This is the idea behind Fano's inequality



Fano's Inequality

- Suppose that we want to estimate the value of a RV X whose distribution is $p(x)$
- We observe a RV Y related to X by the conditional distribution $p(y|x)$
- From Y, we calculate a function $g(Y) = \hat{X}$, where $\hat{X}$ is an estimate of X and takes on values in $\hat{\mathcal{X}}$
- We will not restrict the alphabet $\hat{\mathcal{X}}$ to be equal to $\mathcal{X}$, and we will also allow the function $g(Y)$ to be random
- We wish to bound the probability that $\hat{X} \neq X$
- We observe that $X \rightarrow Y \rightarrow \hat{X}$ forms a Markov chain
- Define the probability of error $P_e = \Pr\{\hat{X} \neq X\}$

Fano's Inequality

- **Theorem:** For any estimator $\hat{X}$ such that $X \to Y \to \hat{X}$, with $P_e = \Pr(X \neq \hat{X})$, we have

$$\underbrace{H_{\mathrm{B}}(P_e)}_{\text{H of the error flag (binary) RV}} + P_e \log|\mathcal{X}| \underbrace{\geq}_{\text{1st}} H\left(X\middle|\hat{X}\right) \underbrace{\geq}_{\text{2nd}} H(X|Y)$$

This inequality can be weakened to

$$1 + P_e \log|\mathcal{X}| \geq H(X|Y) \qquad \text{or} \qquad P_e \geq \frac{H(X|Y) - 1}{\log|\mathcal{X}|}$$

Note that $P_e = 0$ implies that $H(X|Y) = 0$, as intuition suggests

  - *Proof:* We first ignore the role of Y and prove the first inequality. We will then use the data-processing inequality to prove the more traditional form of Fano's inequality, given by the second inequality

Fano's Inequality (proof cont'd)

- Define an «error» random variable $E = \begin{cases} 1 & \hat{X} \neq X \\ 0 & \hat{X} = X \end{cases}$

  Then, use the chain rule for entropies to expand $H(E,X|\hat{X})$

$$H(E,X|\hat{X}) = H(X|\hat{X}) + \underbrace{H(E|X,\hat{X})}_{=0} = \underbrace{H(E|\hat{X})}_{\leq H_B(P_e)} + \underbrace{H(X|E,\hat{X})}_{\leq P_e \log|\mathcal{X}|}$$

  Conditioning reduces entropy, so $H(E|\hat{X}) \leq H(E) = H_B(P_e)$. Since E is a function of X and $\hat{X}$, the conditional $H(E|X,\hat{X})=0$. Also, since E is a binary-valued RV, $H(E)=H_B(P_e)$. The remaining term, $H(X|E,\hat{X})$, can be bounded as

$$\begin{aligned} H(X|E,\hat{X}) &= \Pr(E=0)H(X|\hat{X},E=0) + \Pr(E=1)H(X|\hat{X},E=1) \\ &\leq (1-P_e)0 + P_e \log|\mathcal{X}| \end{aligned}$$

  since given $E=0$, $X=\hat{X}$, and given $E=1$, we can upper bound the conditional entropy by the log of the number of outcomes

- Combining the previous results, we obtain $H_B(P_e) + P_e \log|\mathcal{X}| \geq H(X|\hat{X})$
- By the data-processing inequality, we have $I(X;\hat{X}) \leq I(X;Y)$ since $X \to Y \to \hat{X}$ is a Markov chain, and therefore $H(X|\hat{X}) \geq H(X|Y)$.

  Thus, we have

$$H(P_e) + P_e \log|\mathcal{X}| \geq H(X|\hat{X}) \geq H(X|Y)$$

Fano's Inequality Corollaries

- For any two RVs X and Y, let $p = \Pr(X \neq Y)$. Then

$$H_{\mathrm{B}}(p) + p \log|\mathcal{X}| \geq H(X|Y)$$

  - *Proof:* Let $\hat{X} = Y$ in Fano's inequality

Estimator RV

- Let $P_e = \Pr(X \neq \hat{X})$ and $\hat{X}: \mathcal{Y} \to \mathcal{X}$; then $H_{\mathrm{B}}(P_e) + P_e \log(|\mathcal{X}| - 1) \geq H(X|Y)$
  - *Proof:* change only

$$\begin{aligned} H\left(X \middle| E, \hat{X}\right) &= \Pr(E = 0)\, H\left(X \middle| \hat{X}, E = 0\right) + \Pr(E = 1)\, H\left(X \middle| \hat{X}, E = 1\right) \\ &\leq (1 - P_e)0 + P_e \log(|\mathcal{X}| - 1) \end{aligned}$$

  since given $E = 0$, $X = \hat{X}$, and given $E = 1$, the range of possible X outcomes is $|\mathcal{X}| - 1$, we can upper bound the conditional entropy by $\log(|\mathcal{X}| - 1)$

- For any two random variables X and Y, if the estimator $g(Y)$ takes values in the set $\mathcal{X}$, we can strengthen the inequality slightly by replacing $\log|\mathcal{X}|$ with $\log(|\mathcal{X}| - 1)$

Remark

- Suppose that there is no knowledge of Y
- Thus, X must be guessed without information. Let $X \in \{1,2,\ldots,m\}$ and $p_1 \geq p_2 \geq \ldots \geq p_m$
- Then the best guess of X is $\hat{X} = 1$ and the resulting probability of error is $P_e = 1 - p_1$. Fano's inequality becomes

$$H_{\mathrm{B}}(P_e) + P_e \log(m-1) \geq H(X)$$

- This PMF achieves this bound with equality

$$(p_1, p_2, \ldots, p_m) = \left(1 - P_e, \frac{P_e}{m-1}, \ldots, \frac{P_e}{m-1}\right)$$

p_1 Uniform on the rest

Example: ternary comms

- Consider a ternary communication system transmitting the message $x \in \{0,1,2\}$
- A receiver estimates the transmitted symbol as $y \in \{0,1,2\}$ with a probability of error $P_e = 0.1$

- With Fano we calculate

$$\begin{aligned} H(X|Y) &\leq H_{\mathrm{B}}(P_e) + P_e \log(|\mathcal{X}| - 1) = H_{\mathrm{B}}(0.1) + 0.1 \log(3-1) \\ &= -(0.1 \log 0.1 + 0.9 \log 0.9) + 0.1 \log 2 \\ &= 0.1414 + 0.1 = 0.2414 \end{aligned}$$

- The value of 0.24 bits bounds the remaining uncertainty in the knowledge of *X* given *Y* is received

Lemma

- Let X and X' be two i.i.d. RVs with entropy $H(X)$.
- The probability that $X = X'$ is given by

$$\Pr(X = X') = \sum_{x \in \mathcal{X}} p^2(x) \geq 2^{-H(X)}$$

with equality if and only if X has a uniform distribution

- *Proof:* Suppose that $X \sim p(x)$. By Jensen's inequality and convexity of the function $f(y) = 2^y$, we have

which implies $$2^{\mathrm{E}\{\log p(X)\}} \leq \mathrm{E}\{2^{\log p(X)}\}$$

$$2^{-H(X)} = 2^{\sum_{x \in \mathcal{X}} p(x) \log p(x)} \leq \sum_{x \in \mathcal{X}} p(x) 2^{\log p(x)} = \sum_{x \in \mathcal{X}} p^2(x)$$

Corollary

- Let X,X' be independent with $X \sim p(x)$, $X' \sim r(x)$, x,$x' \in \mathcal{X}$. Then

$$\Pr(X = X') \geq 2^{-H(p)-D(p\|r)} = 2^{-H(p\|r)}$$
$$\Pr(X = X') \geq 2^{-H(r)-D(r\|p)} = 2^{-H(r\|p)}$$

  - *Proof*:

$$2^{-H(p\|r)} = 2^{\sum p(x)\log r(x)} \leq \sum_{x\in\mathcal{X}} p(x)2^{\log r(x)} = \sum_{x\in\mathcal{X}} p(x)r(x) = \Pr(X = X')$$

- Example: in machine learning, the PMF $p(x)$ represents the true distribution of a labeling problem outcomes, and $q(x)$ is the predicted value of the current model
  - The lowest CE is, the highest is the prediction quality: in other terms, the correct labeling probability is maximized